Code No: 138EK **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, June - 2024
REAL-TIME SYSTEMS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	How to create a file?	[2]
b)	What is the difference between wait and waitpid?	[3]
c)	What is a scheduler?	[2]
d)	Distinguish between offline and online scheduling.	[3]
e)	Define the Pipes in RTS.	[2]
f)	Write the function of event register.	[3]
g)	What is an Exception?	[2]
h)	Write the applications of Real Time Clock.	[3]
i)	Write the features of tiny OS.	[2]
j)	Define case studies and give one example with any one of the RTOS.	[3]

PART – B

(50 Marks)

2. What is a file? Explain the different types of I/O files available in UNIX operating System. [10]

OR

3.a) What are the properties that are inherited by a child process from its parent process? Illustrate with example.
b) Write about lseek and write. [5+5]

4. Give a detailed description about the synchronization tools used RTOS. [10]

OR

5.a) What is a semaphore? Explain the states of semaphore.
b) What is the significance of weighted round robin approach for real time scheduling? [5+5]

6.a) Explain RTOS command Shell.
b) Describe about I/O sub system and layered model. [5+5]

OR

7.a) What is a Signal? Explain typical signal operations.
b) What are the various shared data operating system services? [5+5]

8.a) How does an RTOS handle priority inversion and priority inheritance?
b) Discuss about the steps used by the processor for handling an exception. [5+5]

OR

9.a) What do you mean by reentrancy? What are the conditions to be satisfied for a function to be Reentrant?
b) Explain processing of Exceptions and spurious Interrupts. [5+5]

10.a) Explain the data structure of a real time kernel.
b) What is an RT Linux Module? What are the function calls provided for timer management in RT Linux? [5+5]

OR

11. Provide the important features of VxWorks and explain task management functions in VxWorks RTOS. [10]

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